MLOPS Assignment 2

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1. Which run had the lowest latency, and why might that be?

In the latest execution, **run_batch4** had the lowest latency (0.325 s), followed by run_batch8 (0.520 s) and run_batch1 (1.299 s). Batching lets the model process several sentences in one forward pass, so there are fewer inference calls and less per-call overhead (tokenisation, Python and framework overhead). run_batch1 makes 10 separate calls for 10 sentences, so it is the slowest. run_batch4 may also have edged out run_batch8 because it ran last, after the model and caches were already warm, and because 10 samples split into batches of 4 (4+4+2) avoid the padding cost of a larger batch. The exact timings vary a little from execution to execution (in the first execution of this session, run_batch1 took 1.48 s and run_batch8 took 0.621 s), but the trend is the same: larger batches are faster.

2. Did accuracy change between runs? Why or why not?

No. Accuracy stayed at **1.0 (100%)** in every run. All runs used the same pre-trained model (distilbert-base-uncased-finetuned-sst-2-english) with the same weights on the same 10 labelled sentences, and inference is deterministic. Batch size only changes how the inputs are grouped and how fast they are processed, not the model or its predictions.

3. Name two benefits of using MLflow here instead of printing results in the terminal.

• **Reproducibility and record-keeping:** MLflow stores the parameters (model_name, batch_size, num_samples), metrics (accuracy, avg_confidence, total_latency_sec) and artifacts (predictions.csv) of every run permanently. Terminal output is lost once the window is cleared or closed.

• **Easy comparison:** runs can be compared side by side in the UI with tables and charts (parallel coordinates, scatter plots), which is much easier than reading printed lines.

Results summary (latest execution)

Run	batch_size	Accuracy	Latency (s)
run_batch1	1	1.0	1.299
run_batch8	8	1.0	0.520
run_batch4	4	1.0	0.325

(53) WhatsApp | Compare Runs - MLflow | Desktop/mlops-lab/ | mlflow_lab.py

localhost:8888/edit/Desktop/mlops-lab/mlflow_lab.py#

File Edit View Language Python

```python
import time
import mlflow
import pandas as pd
from transformers import pipeline

data = [
    ("I loved this movie, it was fantastic!", "POSITIVE"),
    ("The service was terrible and slow.", "NEGATIVE"),
    ("What a wonderful experience.", "POSITIVE"),
    ("I regret buying this product.", "NEGATIVE"),
    ("The food was delicious and fresh.", "POSITIVE"),
    ("This is the worst app I have ever used.", "NEGATIVE"),
    ("Absolutely brilliant performance by the team.", "POSITIVE"),
    ("The battery dies within an hour.", "NEGATIVE"),
    ("Great value for money, highly recommend.", "POSITIVE"),
    ("I am very disappointed with the quality.", "NEGATIVE"),
]
texts = [t for t, _ in data]
labels = [l for _, l in data]

mlflow.set_experiment("sentiment-inference-lab")

def run_experiment(model_name, batch_size, run_name):
    clf = pipeline("sentiment-analysis", model=model_name)
    with mlflow.start_run(run_name=run_name):
        mlflow.log_param("model_name", model_name)
        mlflow.log_param("batch_size", batch_size)
        mlflow.log_param("num_samples", len(texts))

        start = time.time()
        preds = clf(texts, batch_size=batch_size)
        latency = time.time() - start

        pred_labels = [p["label"] for p in preds]
        accuracy = sum(p == l for p, l in zip(pred_labels, labels)) / len(labels)
        avg_conf = sum(p["score"] for p in preds) / len(preds)

        mlflow.log_metric("accuracy", accuracy)
        mlflow.log_metric("avg_confidence", avg_conf)
        mlflow.log_metric("total_latency_sec", latency)

        df = pd.DataFrame({
            "text": texts,
            "actual": labels,
            "predicted": pred_labels,
            "confidence": [p["score"] for p in preds],
        })
        df.to_csv("predictions.csv", index=False)
        mlflow.log_artifact("predictions.csv")

        print(run_name, "-> accuracy:", accuracy, "latency:", round(latency, 3))

MODEL = "distilbert-base-uncased-finetuned-sst-2-english"
run_experiment(MODEL, batch_size=1, run_name="run_batch1")
run_experiment(MODEL, batch_size=8, run_name="run_batch8")
run_experiment(MODEL, batch_size=4, run_name="run_batch4")
```

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ENG 07:31 29-10-2026

Command Prompt - mlflow ui
22%2512a62294bc954a288f3efb82326ca183%25%22 HTTP/1.1" 200 OK
2026/10/29 07:20:07 INFO: 127.0.0.1:56026 - "POST /ajax-api/3.0/mlflow/ui-telemetry HTTP/1.1" 200 OK
2026/10/29 07:20:15 INFO: 127.0.0.1:56031 - "GET /ajax-api/2.0/mlflow/model-versions/search?filter=tags.%60mlflow.prompt.is_prompt%60+%3D+%27true%27+AND+tags.%60mlflow.prompt.run_ids%60+ILIKE+%
22%2512a62294bc954a288f3efb82326ca183%25%22 HTTP/1.1" 200 OK
2026/10/29 07:20:16 INFO: 127.0.0.1:56032 - "GET /static-files/static/css/9288.ecc3b598.chunk.css HTTP/1.1" 200 OK
2026/10/29 07:20:16 INFO: 127.0.0.1:56034 - "GET /static-files/static/js/2825.b62ad5a7.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:20:16 INFO: 127.0.0.1:56033 - "GET /static-files/static/js/9288.2ca2efc9.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:20:16 INFO: 127.0.0.1:56031 - "GET /static-files/static/js/4463.58bdf1f4.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:20:33 INFO: 127.0.0.1:56038 - "POST /ajax-api/2.0/mlflow/experiments/search-datasets HTTP/1.1" 200 OK
2026/10/29 07:20:33 INFO: 127.0.0.1:56039 - "GET /ajax-api/2.0/mlflow/gateway-proxy?gateway_path=api%2F2.0%2Fendpoints%2F HTTP/1.1" 200 OK
2026/10/29 07:20:33 INFO: 127.0.0.1:56038 - "POST /ajax-api/2.0/mlflow/logged-models/search HTTP/1.1" 200 OK
2026/10/29 07:20:33 INFO: 127.0.0.1:56040 - "POST /ajax-api/2.0/mlflow/runs/search HTTP/1.1" 200 OK
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2026/10/29 07:20:35 INFO: 127.0.0.1:56038 - "GET /ajax-api/2.0/mlflow/model-versions/search?filter=tags.%60mlflow.prompt.is_prompt%60+%3D+%27true%27+AND+tags.%60mlflow.prompt.run_ids%60+ILIKE+%
22%25b11a21106a7449c4af8de83ed130f5bd%25%22 HTTP/1.1" 200 OK
2026/10/29 07:20:35 INFO: 127.0.0.1:56040 - "POST /ajax-api/2.0/mlflow/runs/search HTTP/1.1" 200 OK
2026/10/29 07:20:37 INFO: 127.0.0.1:56040 - "POST /ajax-api/3.0/mlflow/ui-telemetry HTTP/1.1" 200 OK
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22%25b11a21106a7449c4af8de83ed130f5bd%25%22 HTTP/1.1" 200 OK
2026/10/29 07:21:06 INFO: 127.0.0.1:56046 - "POST /ajax-api/2.0/mlflow/experiments/search-datasets HTTP/1.1" 200 OK
2026/10/29 07:21:06 INFO: 127.0.0.1:56047 - "GET /ajax-api/2.0/mlflow/gateway-proxy?gateway_path=api%2F2.0%2Fendpoints%2F HTTP/1.1" 200 OK
2026/10/29 07:21:06 INFO: 127.0.0.1:56046 - "POST /ajax-api/2.0/mlflow/logged-models/search HTTP/1.1" 200 OK
2026/10/29 07:21:06 INFO: 127.0.0.1:56048 - "POST /ajax-api/2.0/mlflow/runs/search HTTP/1.1" 200 OK
2026/10/29 07:21:07 INFO: 127.0.0.1:56046 - "GET /ajax-api/2.0/mlflow/model-versions/search?filter=run_id%3D%27f3441df224db4b98817624a80d52b832%27 HTTP/1.1" 200 OK
2026/10/29 07:21:07 INFO: 127.0.0.1:56048 - "POST /graphql HTTP/1.1" 200 OK
2026/10/29 07:21:07 INFO: 127.0.0.1:56046 - "GET /ajax-api/2.0/mlflow/model-versions/search?filter=tags.%60mlflow.prompt.is_prompt%60+%3D+%27true%27+AND+tags.%60mlflow.prompt.run_ids%60+ILIKE+%
22%25f3441df224db4b98817624a80d52b832%25%22 HTTP/1.1" 200 OK
2026/10/29 07:21:07 INFO: 127.0.0.1:56048 - "POST /ajax-api/2.0/mlflow/runs/search HTTP/1.1" 200 OK
2026/10/29 07:21:08 INFO: 127.0.0.1:56048 - "POST /ajax-api/3.0/mlflow/ui-telemetry HTTP/1.1" 200 OK
2026/10/29 07:21:17 INFO: 127.0.0.1:56053 - "GET /ajax-api/2.0/mlflow/model-versions/search?filter=tags.%60mlflow.prompt.is_prompt%60+%3D+%27true%27+AND+tags.%60mlflow.prompt.run_ids%60+ILIKE+%
22%25f3441df224db4b98817624a80d52b832%25%22 HTTP/1.1" 200 OK
2026/10/29 07:21:26 INFO: 127.0.0.1:56055 - "GET /ajax-api/2.0/mlflow/model-versions/search?filter=tags.%60mlflow.prompt.is_prompt%60+%3D+%27true%27+AND+tags.%60mlflow.prompt.run_ids%60+ILIKE+%
22%25f3441df224db4b98817624a80d52b832%25%22 HTTP/1.1" 200 OK
2026/10/29 07:21:37 INFO: 127.0.0.1:56057 - "POST /ajax-api/3.0/mlflow/ui-telemetry HTTP/1.1" 200 OK
2026/10/29 07:21:50 INFO: 127.0.0.1:56060 - "POST /ajax-api/2.0/mlflow/experiments/search-datasets HTTP/1.1" 200 OK
2026/10/29 07:21:50 INFO: 127.0.0.1:56062 - "GET /ajax-api/2.0/mlflow/gateway-proxy?gateway_path=api%2F2.0%2Fendpoints%2F HTTP/1.1" 200 OK
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2026/10/29 07:21:50 INFO: 127.0.0.1:56061 - "POST /ajax-api/2.0/mlflow/runs/search HTTP/1.1" 200 OK
2026/10/29 07:21:53 INFO: 127.0.0.1:56062 - "GET /static-files/static/css/1376.dda5119a.chunk.css HTTP/1.1" 200 OK
2026/10/29 07:21:53 INFO: 127.0.0.1:56060 - "GET /static-files/static/js/1886.c80613f9.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:21:53 INFO: 127.0.0.1:56064 - "GET /static-files/static/js/1376.af1668fb.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:21:53 INFO: 127.0.0.1:56061 - "GET /static-files/static/js/6015.1659bdf2.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:21:53 INFO: 127.0.0.1:56061 - "GET /ajax-api/2.0/mlflow/experiments/get?experiment_id=1 HTTP/1.1" 200 OK
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2026/10/29 07:22:07 INFO: 127.0.0.1:56065 - "POST /ajax-api/3.0/mlflow/ui-telemetry HTTP/1.1" 200 OK
2026/10/29 07:23:07 INFO: 127.0.0.1:56067 - "POST /ajax-api/3.0/mlflow/ui-telemetry HTTP/1.1" 200 OK
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Command Prompt - mlflow ui

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2026/10/29 07:17:08 INFO:     127.0.0.1:55971 - "GET /static-files/static/media/demo-tracing-screenshot.f95794fa205aa8e6c386.png HTTP/1.1" 200 OK
2026/10/29 07:17:08 INFO:     127.0.0.1:55975 - "GET /ajax-api/3.0/mlflow/assistant/providers HTTP/1.1" 200 OK
2026/10/29 07:17:08 INFO:     127.0.0.1:55972 - "GET /ajax-api/2.0/mlflow/users/current HTTP/1.1" 404 Not Found
2026/10/29 07:17:08 INFO:     127.0.0.1:55976 - "GET /static-files/static/css/5027.26533251.chunk.css HTTP/1.1" 200 OK
2026/10/29 07:17:08 INFO:     127.0.0.1:55977 - "GET /static-files/static/js/5027.a66ffe35.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:17:08 INFO:     127.0.0.1:55971 - "GET /ajax-api/3.0/mlflow/assistant/providers HTTP/1.1" 200 OK
2026/10/29 07:18:01 INFO:     127.0.0.1:55990 - "GET /ajax-api/2.0/mlflow/experiments/search?max_results=5&order_by=last_update_time+DESC HTTP/1.1" 200 OK
2026/10/29 07:19:05 INFO:     127.0.0.1:55998 - "GET /ajax-api/2.0/mlflow/experiments/search?max_results=5&order_by=last_update_time+DESC HTTP/1.1" 200 OK
2026/10/29 07:19:22 INFO:     127.0.0.1:56003 - "GET /static-files/static/js/2809.4115baa4.chunk.js HTTP/1.1" 200 OK
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2026/10/29 07:19:22 INFO:     127.0.0.1:56002 - "GET /static-files/static/js/3448.da9a79ba.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:22 INFO:     127.0.0.1:56000 - "GET /static-files/static/js/6507.352ad2e0.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:22 INFO:     127.0.0.1:56004 - "GET /static-files/static/js/5439.b5816bbb.chunk.js HTTP/1.1" 200 OK
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2026/10/29 07:19:22 INFO:     127.0.0.1:56001 - "GET /static-files/static/js/353.df7e2039.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:22 INFO:     127.0.0.1:56003 - "POST /graphql HTTP/1.1" 200 OK
2026/10/29 07:19:23 INFO:     127.0.0.1:56004 - "GET /static-files/static/js/9848.664e1dcc.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:23 INFO:     127.0.0.1:56003 - "GET /static-files/static/js/3198.bb9ac57b.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:23 INFO:     127.0.0.1:56004 - "GET /static-files/static/css/5343.b4cc2aa3.chunk.css HTTP/1.1" 200 OK
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2026/10/29 07:19:23 INFO:     127.0.0.1:55999 - "GET /static-files/static/js/8503.9d34794c.chunk.js HTTP/1.1" 200 OK
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2026/10/29 07:19:23 INFO:     127.0.0.1:56004 - "GET /static-files/static/js/9360.ea3236af.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:23 INFO:     127.0.0.1:56002 - "GET /static-files/static/js/2369.90891e4f.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:23 INFO:     127.0.0.1:56000 - "GET /static-files/static/js/3121.3be60de2.chunk.js HTTP/1.1" 200 OK
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2026/10/29 07:19:23 INFO:     127.0.0.1:56004 - "GET /static-files/static/js/3793.0117dc93.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:23 INFO:     127.0.0.1:56001 - "GET /static-files/static/js/4751.94bc6db7.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:23 INFO:     127.0.0.1:56002 - "GET /static-files/static/js/9662.9a8db43d.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:23 INFO:     127.0.0.1:56000 - "GET /static-files/static/js/6475.82f59d0e.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:23 INFO:     127.0.0.1:56002 - "GET /static-files/static/js/5343.bbb1f624.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:23 INFO:     127.0.0.1:55999 - "GET /static-files/static/js/5702.205ea7af.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:23 INFO:     127.0.0.1:56004 - "GET /static-files/static/js/3878.0467b2c4.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:23 INFO:     127.0.0.1:56004 - "POST /ajax-api/2.0/mlflow/experiments/search-datasets HTTP/1.1" 200 OK
2026/10/29 07:19:23 INFO:     127.0.0.1:56004 - "GET /static-files/static/css/2685.f073c5c3.chunk.css HTTP/1.1" 200 OK
2026/10/29 07:19:23 INFO:     127.0.0.1:56004 - "GET /ajax-api/2.0/mlflow/gateway-proxy?gateway_path=api%2F2.0%2Fendpoints%2F HTTP/1.1" 200 OK
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2026/10/29 07:19:23 INFO:     127.0.0.1:56003 - "POST /ajax-api/2.0/mlflow/runs/search HTTP/1.1" 200 OK
2026/10/29 07:19:23 INFO:     127.0.0.1:56002 - "POST /ajax-api/2.0/mlflow/logged-models/search HTTP/1.1" 200 OK
2026/10/29 07:19:24 INFO:     127.0.0.1:55999 - "GET /static-files/static/js/2685.2f8f1b38.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:38 INFO:     127.0.0.1:56010 - "POST /ajax-api/3.0/mlflow/ui-telemetry HTTP/1.1" 200 OK
2026/10/29 07:19:43 INFO:     127.0.0.1:56023 - "GET /static-files/static/css/2684.13093132.chunk.css HTTP/1.1" 200 OK
2026/10/29 07:19:43 INFO:     127.0.0.1:56018 - "GET /static-files/static/js/930.7554884c.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:43 INFO:     127.0.0.1:56019 - "GET /static-files/static/js/7720.1cbd9501.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:43 INFO:     127.0.0.1:56022 - "GET /static-files/static/js/3254.9017d695.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:43 INFO:     127.0.0.1:56021 - "GET /static-files/static/js/5995.b8faaf19.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:43 INFO:     127.0.0.1:56020 - "GET /static-files/static/js/4649.e79cc735.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:43 INFO:     127.0.0.1:56023 - "GET /static-files/static/js/7349.715264bd.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:43 INFO:     127.0.0.1:56018 - "GET /static-files/static/js/9930.05913dd5.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:19:44 INFO:     127.0.0.1:56019 - "GET /ajax-api/2.0/mlflow/model-versions/search?filter=run_id%3D%2712a62294bc954a288f3efb82326ca183%27 HTTP/1.1" 200 OK
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(53) WhatsApp | Compare Runs - MLflow

127.0.0.1:5000/#/compare-runs?runs=["12a62294bc954a288f3efb82326ca183","b11a21106a7449c4af8de83ed130f5bd","f3441df224db4b98817624a80d52b832"]&experiments=["1"]

mlflow 3.17.2

GenAI | Model training

sentiment-inferen...

Runs

Models

Traces

Model registry

Docs

Settings

Run details

Run ID:	12a62294bc954a288f3efb82326ca183	b11a21106a7449c4af8de83ed130f5bd	f3441df224db4b98817624a80d52b832
Run Name:	run_batch4	run_batch8	run_batch1
Start Time:	10/29/2026, 07:15:57 AM	10/29/2026, 07:15:55 AM	10/29/2026, 07:15:53 AM
End Time:	10/29/2026, 07:15:57 AM	10/29/2026, 07:15:55 AM	10/29/2026, 07:15:53 AM
Duration:	138ms	134ms	215ms

Parameters

Show diff only

batch_size	4	8	1
model_name	distilbert-base-uncased-finetuned-sst-2-english	distilbert-base-uncased-finetuned-sst-2-english	distilbert-base-uncased-finetuned-sst-2-english
num_samples	10	10	10

Metrics

Show diff only

accuracy	1	1	1
avg_confidence	1	1	1
total_latency_sec	0.074	0.071	0.139

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ENG 07:23

mlflow 3.17.2

GenAI | Model training

sentiment-inferen...

- Runs
- Models
- Traces
- Model registry

sentiment-inference-lab ›

Comparing 3 Runs from 1 Experiment

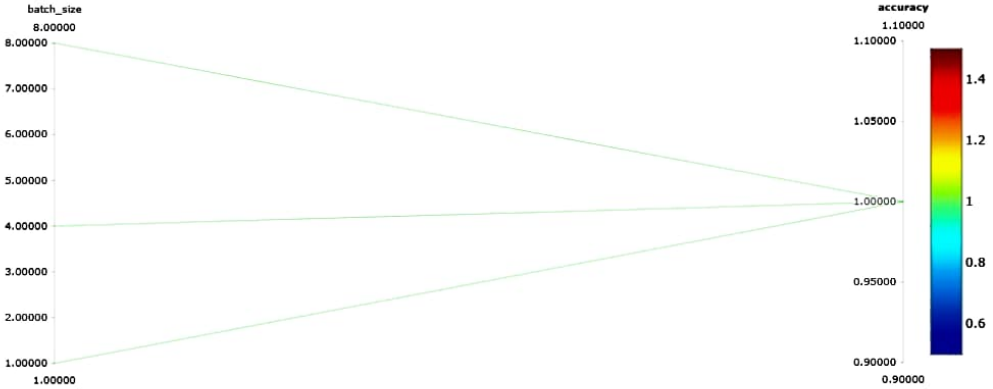
Visualizations

Parallel Coordinates Plot | Scatter Plot | Box Plot | Contour Plot

Parameters: batch_size ×

Metrics: accuracy ×

Clear All



Run details

Run ID: 12a62294bc954a288f3efb82326ca183 | b11a21106a7449c4af8de83ed130f5bd | f3441df224db4b98817624a80d52b832

Docs

Settings

(53) WhatsApp | Compare Runs - MLflow

127.0.0.1:5000/#/compare-runs?runs=["12a62294bc954a288f3efb82326ca183","b11a21106a7449c4af8de83ed130f5bd","f3441df224db4b98817624a80d52b832"]&experiments=["1"]

mlflow 3.17.2

GenAI | Model training

sentiment-inferen...

Runs
Models
Traces

Model registry

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sentiment-inference-lab ›

Comparing 3 Runs from 1 Experiment

Visualizations

Parallel Coordinates Plot | Scatter Plot | Box Plot | Contour Plot

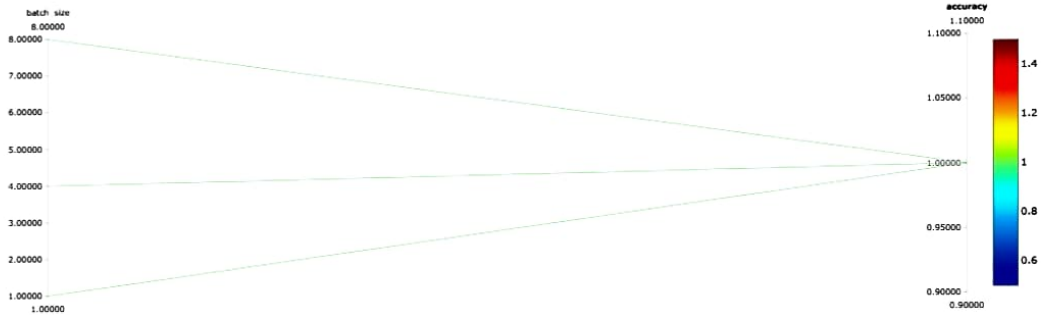
Parameters:

batch_size ✕

Metrics:

accuracy ✕

Clear All

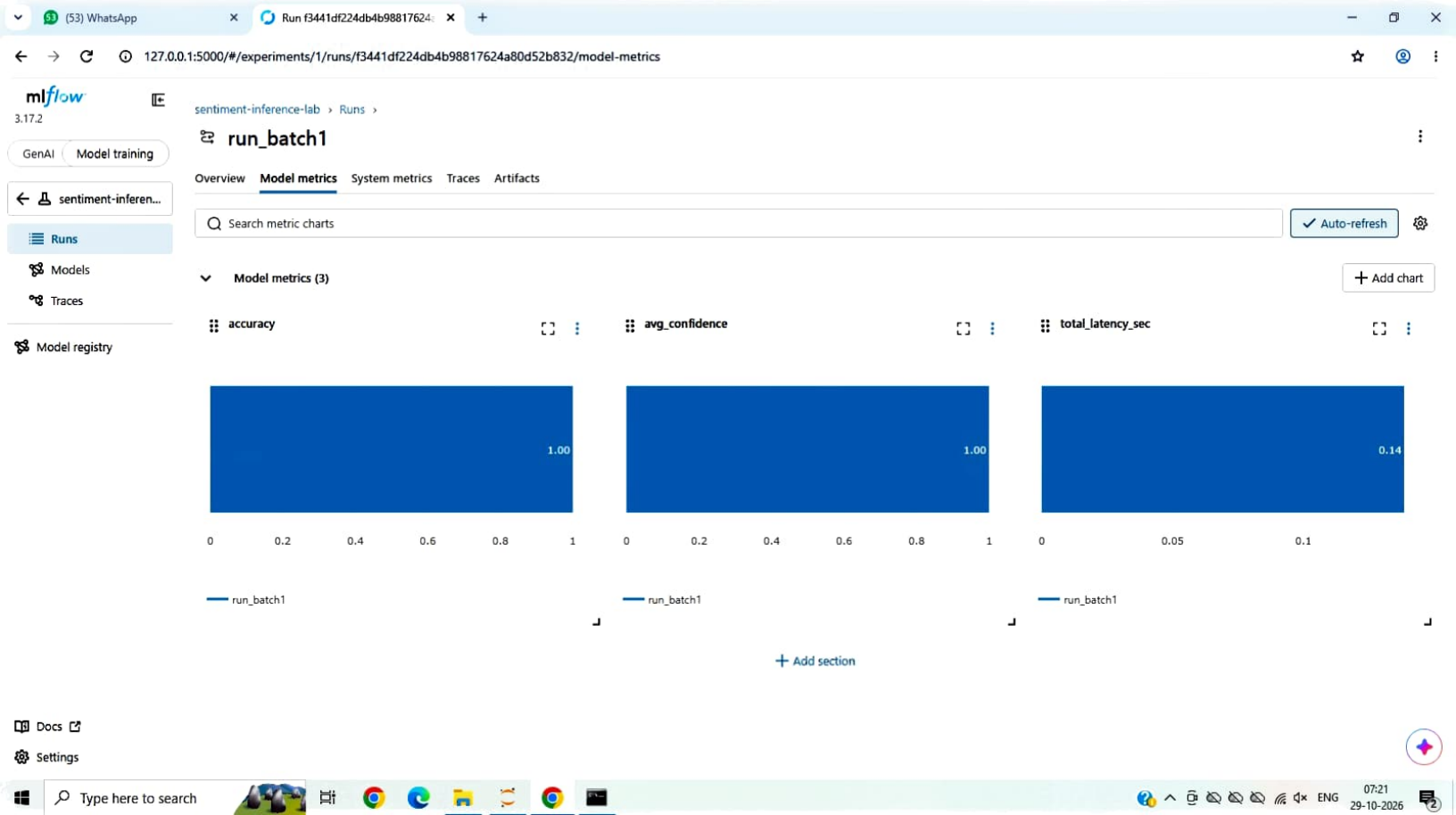


Run details

Run ID:	12a62294bc954a288f3efb82326ca183	b11a21106a7449c4af8de83ed130f5bd	f3441df224db4b98817624a80d52b832
Run Name:	run_batch4	run_batch8	run_batch1
Start Time:	10/29/2026, 07:15:57 AM	10/29/2026, 07:15:55 AM	10/29/2026, 07:15:53 AM
End Time:	10/29/2026, 07:15:57 AM	10/29/2026, 07:15:55 AM	10/29/2026, 07:15:53 AM
Duration:	138ms	134ms	215ms

Type here to search

ENG 07:22 29-10-2026


(53) WhatsApp
Run f3441df224db4b98817624
127.0.0.1:5000/#/experiments/1/runs/f3441df224db4b98817624a80d52b832/model-metrics
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
sentiment-inference-lab › Runs ›
run_batch1
Overview
Model metrics
System metrics
Traces
Artifacts
Search metric charts
Auto-refresh
Model metrics (3)
Add chart
accuracy
1.00
0
0.2
0.4
0.6
0.8
1
run_batch1
avg_confidence
1.00
0
0.2
0.4
0.6
0.8
1
run_batch1
total_latency_sec
0.14
0
0.05
0.1
run_batch1
Add section
Docs
Settings
Type here to search
ENG
07:21
29-10-2026

mlflow
3.17.2

GenAI | Model training

sentiment-inferen...

- Runs
- Models
- Traces

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sentiment-inference-lab › Runs ›

run_batch1

Overview | Model metrics | System metrics | Traces | Artifacts

Description

No description

Metrics (3)

Search metrics

Metric	Value
accuracy	1
avg_confidence	0.9996546626091003
total_latency_sec	0.13858485221862793

Parameters (3)

Search parameters

Parameter	Value
model_name	distilbert-base-uncased-finetuned-sst-2-english
batch_size	1
num_samples	10

About this run

Created at	10/29/2026, 07:15:53 AM
Created by	GSSS
Experiment ID	1
Status	Finished
Run ID	f3441df224db4b98817624a80d52b832
Duration	215ms
Source	mlflow_lab.py
Logged models	—
Registered prompts	—

Datasets

None

Tags

Add tags

Registered models

None

mlflow
3.17.2

GenAI | Model training

sentiment-inferen...

- Runs
- Models
- Traces

Model registry

sentiment-inference-lab › Runs ›

run_batch8

Overview | **Model metrics** | System metrics | Traces | Artifacts

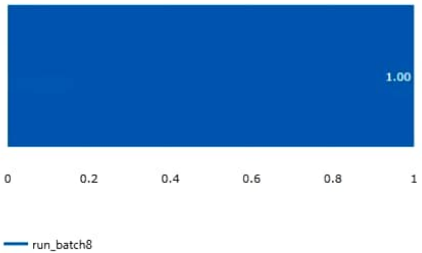
Search metric charts

Auto-refresh

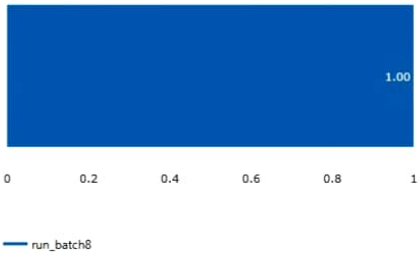
Model metrics (3)

+ Add chart

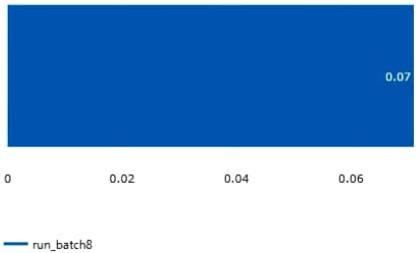
accuracy



avg_confidence



total_latency_sec



+ Add section

Docs
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3.17.2

GenAI | Model training

sentiment-inferen...

- Runs
- Models
- Traces

Model registry

Docs
Settings

sentiment-inference-lab › Runs ›

run_batch8

Overview | Model metrics | System metrics | Traces | Artifacts

Description

No description

Metrics (3)

Search metrics

Metric	Value
accuracy	1
avg_confidence	0.9996546626091003
total_latency_sec	0.07104682922363281

Parameters (3)

Search parameters

Parameter	Value
model_name	distilbert-base-uncased-finetuned-sst-2-english
batch_size	8
num_samples	10

About this run

Created at	10/29/2026, 07:15:55 AM
Created by	GSSS
Experiment ID	1
Status	Finished
Run ID	b11a21106a7449c4af8de83ed130f5bd
Duration	134ms
Source	mlflow_lab.py
Logged models	—
Registered prompts	—

Datasets

None

Tags

Add tags

Registered models

None

mlflow
3.17.2

GenAI | Model training

sentiment-inferen...

- Runs
- Models
- Traces

Model registry

Docs
Settings

sentiment-inference-lab › Runs ›

run_batch4

Overview | Model metrics | System metrics | Traces | Artifacts

Search metric charts

Auto-refresh

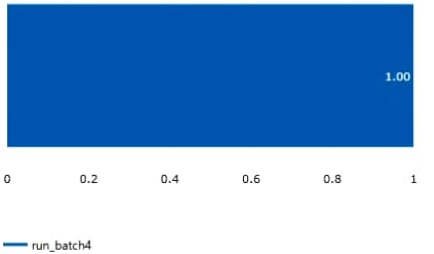
Model metrics (3)

Add chart

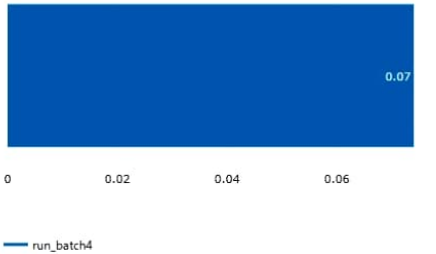
accuracy



avg_confidence



total_latency_sec



Add section

mlflow
3.17.2

GenAI | Model training

sentiment-inferen...

- Runs
- Models
- Traces

Model registry

Docs
Settings

sentiment-inference-lab › Runs ›

run_batch4

Overview | Model metrics | System metrics | Traces | Artifacts

Description

No description

Metrics (3)

Search metrics

Metric	Value
accuracy	1
avg_confidence	0.9996546626091003
total_latency_sec	0.07406115531921387

Parameters (3)

Search parameters

Parameter	Value
model_name	distilbert-base-uncased-finetuned-sst-2-english
batch_size	4
num_samples	10

About this run

Created at	10/29/2026, 07:15:57 AM
Created by	GSSS
Experiment ID	1
Status	Finished
Run ID	12a62294bc954a288f3efb82326ca183
Duration	138ms
Source	mlflow_lab.py
Logged models	—
Registered prompts	—

Datasets

None

Tags

Add tags

Registered models

None

(53) WhatsApp | Runs - Experiment 1 - MLflow

127.0.0.1:5000/#/experiments/1/runs?workflowType=machine_learning&searchFilter=&orderByKey=attributes.start_time&orderByAsc=false&startTime=ALL&lifecycleFilter=Active&modelVersionFilter=All+Runs&...

mlflow 3.17.2

GenAI | Model training

sentiment-inferen...

- Runs
- Models
- Traces

Model registry

Docs

Settings

Experiments › sentiment-inference-lab ›

Runs

Views

metrics.rmse < 1 and params.model = "tree"

All time

New run

State: Active | Datasets | Sort: Created | Columns | Group by

Run Name	Created	Dataset	Duration	Source	Mode
run_batch4	3 minutes ago	-	138ms	mlflow_l...	-
run_batch8	3 minutes ago	-	134ms	mlflow_l...	-
run_batch1	3 minutes ago	-	215ms	mlflow_l...	-

3 matching runs

MLflow Assistant Beta

Ask questions about your experiments, traces, evaluations, and more.

- Compare the selected runs.
- Which run performed best?
- What parameters differ between runs?

Ask a question...

Ollama

Type here to search

ENG 07:19 29-10-2026

Command Prompt - mlflow ui

```
Registry store URI not provided. Using backend store URI.
[MLflow] Security middleware enabled with default settings (localhost-only). To allow connections from other hosts, use --host 0.0.0.0 and configure --allowed-hosts and --cors-allowed-origins.
2026/10/29 07:16:29 WARNING mlflow.server: MLflow job execution requirements not met (MLflow job backend does not support Windows system.). Server will start without job execution support. Errors w
ill be surfaced at job invocation time.
2026/10/29 07:16:30 INFO:     Uvicorn running on http://127.0.0.1:5000 (Press CTRL+C to quit)
2026/10/29 07:16:30 INFO:     Started parent process [15456]
C:\Users\GSSS\Desktop\mlops-lab\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future rel
ease. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\GSSS\Desktop\mlops-lab\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future rel
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  from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\GSSS\Desktop\mlops-lab\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future rel
ease. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\GSSS\Desktop\mlops-lab\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future rel
ease. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_environ
2026/10/29 07:16:32 INFO:     Started server process [7712]
2026/10/29 07:16:32 INFO:     Started server process [17020]
2026/10/29 07:16:32 INFO:     Waiting for application startup.
2026/10/29 07:16:32 INFO:     Waiting for application startup.
2026/10/29 07:16:32 INFO:     Started server process [10948]
2026/10/29 07:16:32 INFO:     Started server process [1580]
2026/10/29 07:16:32 INFO:     Waiting for application startup.
2026/10/29 07:16:32 INFO:     Waiting for application startup.
2026/10/29 07:16:32 INFO:     Application startup complete.
2026/10/29 07:16:32 INFO:     Application startup complete.
2026/10/29 07:16:32 INFO:     Application startup complete.
2026/10/29 07:16:32 INFO:     Application startup complete.
2026/10/29 07:17:07 INFO:     127.0.0.1:55971 - "GET / HTTP/1.1" 200 OK
2026/10/29 07:17:07 INFO:     127.0.0.1:55973 - "GET /static-files/manifest.json HTTP/1.1" 200 OK
2026/10/29 07:17:07 INFO:     127.0.0.1:55971 - "GET /static-files/static/js/main.c2f5f8d4.js HTTP/1.1" 200 OK
2026/10/29 07:17:07 INFO:     127.0.0.1:55972 - "GET /static-files/static/css/main.b0f58ef9.css HTTP/1.1" 200 OK
2026/10/29 07:17:07 INFO:     127.0.0.1:55971 - "GET /static-files/TelemetryLogger.telemetry-worker.d0f8ea2c83c08b63528f.worker.js HTTP/1.1" 200 OK
2026/10/29 07:17:07 INFO:     127.0.0.1:55972 - "GET /static-files/favicon.ico HTTP/1.1" 200 OK
2026/10/29 07:17:07 INFO:     127.0.0.1:55971 - "GET /ajax-api/3.0/mlflow/server-info HTTP/1.1" 200 OK
2026/10/29 07:17:08 INFO:     127.0.0.1:55971 - "GET /static-files/static/js/4205.9e750e7b.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:17:08 INFO:     127.0.0.1:55976 - "GET /ajax-api/3.0/mlflow/assistant/config HTTP/1.1" 200 OK
2026/10/29 07:17:08 INFO:     127.0.0.1:55976 - "GET /ajax-api/3.0/mlflow/assistant/config HTTP/1.1" 200 OK
2026/10/29 07:17:08 INFO:     127.0.0.1:55971 - "GET /static-files/static/js/3877.9df04fb4.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:17:08 INFO:     127.0.0.1:55973 - "GET /ajax-api/3.0/mlflow/ui-telemetry HTTP/1.1" 200 OK
2026/10/29 07:17:08 INFO:     127.0.0.1:55971 - "GET /static-files/static/js/7340.cdf02082.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:17:08 INFO:     127.0.0.1:55973 - "GET /ajax-api/2.0/mlflow/experiments/search?max_results=5&order_by=last_update_time+DESC HTTP/1.1" 200 OK
2026/10/29 07:17:08 INFO:     127.0.0.1:55971 - "GET /static-files/static/media/demo-tracing-screenshot.f95794fa205aa8e6c386.png HTTP/1.1" 200 OK
2026/10/29 07:17:08 INFO:     127.0.0.1:55975 - "GET /ajax-api/3.0/mlflow/assistant/providers HTTP/1.1" 200 OK
2026/10/29 07:17:08 INFO:     127.0.0.1:55972 - "GET /ajax-api/2.0/mlflow/users/current HTTP/1.1" 404 Not Found
2026/10/29 07:17:08 INFO:     127.0.0.1:55976 - "GET /static-files/static/css/5027.26533251.chunk.css HTTP/1.1" 200 OK
2026/10/29 07:17:08 INFO:     127.0.0.1:55977 - "GET /static-files/static/js/5027.a66ffe35.chunk.js HTTP/1.1" 200 OK
2026/10/29 07:17:08 INFO:     127.0.0.1:55971 - "GET /ajax-api/3.0/mlflow/assistant/providers HTTP/1.1" 200 OK
2026/10/29 07:18:01 INFO:     127.0.0.1:55990 - "GET /ajax-api/2.0/mlflow/experiments/search?max_results=5&order_by=last_update_time+DESC HTTP/1.1" 200 OK
```

Type here to search
ENG
07:18
29-10-2026

Command Prompt - mlflow ui

```
d files but your machine does not support them in C:\Users\GSSS\.cache\huggingface\hub\models--distilbert-base-uncased-finetuned-sst-2-english. Caching files will still work but in a degraded versi
on that might require more space on your disk. This warning can be disabled by setting the `HF_HUB_DISABLE_SYMLINKS_WARNING` environment variable. For more details, see https://huggingface.co/docs/
huggingface_hub/how-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer Mode or to run Python as an administrator. In order to activate developer mode, see this article: https://docs.microsoft.com/en
-us/windows/apps/get-started/enable-your-device-for-development
  warnings.warn(message)
C:\Users\GSSS\Desktop\mlops-lab\mlops-env\Lib\site-packages\huggingface_hub\file_download.py:149: UserWarning: `huggingface_hub` cache-system uses symlinks by default to efficiently store duplicate
d files but your machine does not support them in C:\Users\GSSS\.cache\huggingface\hub. Caching files will still work but in a degraded version that might require more space on your disk. This warn
ing can be disabled by setting the `HF_HUB_DISABLE_SYMLINKS_WARNING` environment variable. For more details, see https://huggingface.co/docs/huggingface_hub/how-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer Mode or to run Python as an administrator. In order to activate developer mode, see this article: https://docs.microsoft.com/en
-us/windows/apps/get-started/enable-your-device-for-development
  warnings.warn(message)
model.safetensors: downloading bytes: |                                                                                | 250MB, 18.9MB/s
model.safetensors: reconstructing file: 100%|                                                                          | 268MB /  268MB, 22.8MB/s
Loading weights: 100%|                                                                                                 | 104/104 [00:00<00:00, 1339.98it/s]
tokenizer_config.json: 100%|                                                                                           | 48.0/48.0 [00:00<00:00, 52.2kB/s]
vocab.txt: 100%|                                                                                                       | 232k/232k [00:00<00:00, 4.94MB/s]

(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>notepad mlflow_lab.py

(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>python mlflow_lab.py
2026/10/29 07:15:50 INFO mlflow.store.db.utils: Creating initial MLflow database tables...
2026/10/29 07:15:50 INFO mlflow.store.db.utils: Updating database tables
2026/10/29 07:15:51 INFO mlflow.tracking.fluent: Experiment with name 'sentiment-inference-lab' does not exist. Creating a new experiment.
Loading weights: 100%|                                                                                                 | 104/104 [00:00<00:00, 1615.67it/s]
run_batch1 -> accuracy: 1.0 latency: 0.139
Loading weights: 100%|                                                                                                 | 104/104 [00:00<00:00, 1102.13it/s]
run_batch8 -> accuracy: 1.0 latency: 0.071
Loading weights: 100%|                                                                                                 | 104/104 [00:00<00:00, 1286.99it/s]
run_batch4 -> accuracy: 1.0 latency: 0.074

(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>mlflow ui
Backend store URI not provided. Using sqlite:///mlflow.db
Registry store URI not provided. Using backend store URI.
[MLflow] Security middleware enabled with default settings (localhost-only). To allow connections from other hosts, use --host 0.0.0.0 and configure --allowed-hosts and --cors-allowed-origins.
2026/10/29 07:16:29 WARNING mlflow.server: MLflow job execution requirements not met (MLflow job backend does not support Windows system.). Server will start without job execution support. Errors w
ill be surfaced at job invocation time.
2026/10/29 07:16:30 INFO:     Uvicorn running on http://127.0.0.1:5000 (Press CTRL+C to quit)
2026/10/29 07:16:30 INFO:     Started parent process [15456]
C:\Users\GSSS\Desktop\mlops-lab\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future rel
ease. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\GSSS\Desktop\mlops-lab\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future rel
ease. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\GSSS\Desktop\mlops-lab\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future rel
ease. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\GSSS\Desktop\mlops-lab\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future rel
ease. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_environ
2026/10/29 07:16:32 INFO:     Started server process [7712]
```

Type here to search
ENG
07:18
29-10-2026

mlflow_lab.py file:

```
import time
import mlflow
import pandas as pd
from transformers import pipeline

# Small labelled dataset (10 sentences)
data = [
    ("I loved this movie, it was fantastic!", "POSITIVE"),
    ("The service was terrible and slow.", "NEGATIVE"),
    ("What a wonderful experience.", "POSITIVE"),
    ("I regret buying this product.", "NEGATIVE"),
    ("The food was delicious and fresh.", "POSITIVE"),
    ("This is the worst app I have ever used.", "NEGATIVE"),
    ("Absolutely brilliant performance by the team.", "POSITIVE"),
    ("The battery dies within an hour.", "NEGATIVE"),
    ("Great value for money, highly recommend.", "POSITIVE"),
    ("I am very disappointed with the quality.", "NEGATIVE"),
]

texts = [text for text, _ in data]
labels = [label for _, label in data]

mlflow.set_experiment("sentiment-inference-lab")

def run_experiment(model_name, batch_size, run_name):
    clf = pipeline("sentiment-analysis", model=model_name)

    with mlflow.start_run(run_name=run_name):
        # Parameters
        mlflow.log_param("model_name", model_name)
        mlflow.log_param("batch_size", batch_size)
        mlflow.log_param("num_samples", len(texts))

        # Inference
        start = time.time()
        preds = clf(texts, batch_size=batch_size)
        latency = time.time() - start

        pred_labels = [prediction["label"] for prediction in preds]
        accuracy = sum(predicted == actual for predicted, actual in zip(pred_labels, labels)) / len(labels)
        avg_confidence = sum(prediction["score"] for prediction in preds) / len(preds)

        # Metrics
        mlflow.log_metric("accuracy", accuracy)
        mlflow.log_metric("avg_confidence", avg_confidence)
        mlflow.log_metric("total_latency_sec", latency)

        # Artifact
        predictions =
            pd.DataFrame({ "text":
            texts,
            "actual": labels,
            "predicted": pred_labels,
            "confidence": [prediction["score"] for prediction in preds],
        })
        predictions.to_csv("predictions.csv", index=False)
        mlflow.log_artifact("predictions.csv")

        print(run_name, "-> accuracy:", accuracy, "latency:", round(latency, 3))

MODEL = "distilbert-base-uncased-finetuned-sst-2-english"
run_experiment(MODEL, batch_size=1, run_name="run_batch1")
run_experiment(MODEL, batch_size=8, run_name="run_batch8")
run_experiment(MODEL, batch_size=4, run_name="run_batch4")
```